UNIT 9

MAGNETISM

Learning objectives:

- Learn about magnets.
- Magnetic and nonmagnetic material.
- Properties of magnetic material.
- Use of magnets in daily life.

9.1 What is Magnet?

Objects which attract magnetic materials like cobalt, nickel and iron are called a magnet.

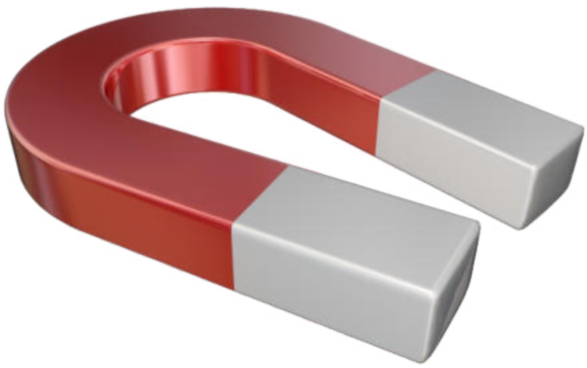
9.2 History of Magnets

The magnets obtained naturally from a Magnetite rock are called natural magnets, and those magnets prepared by the combination of certain mineral ores are called artificial magnets.

There are two types of material.

1. Magnetic Materials: Cobalt, nickel, and iron are some examples of Magnetic Materials. These materials are easily attracted by a magnet.

2. Non-magnetic Materials: Aluminium, zinc, wood, and rubber are called Non-Magnetic Materials, as these materials are not attracted towards the magnet even when they are brought closer to the magnet.



Magnetic and Non-magnetic Materials

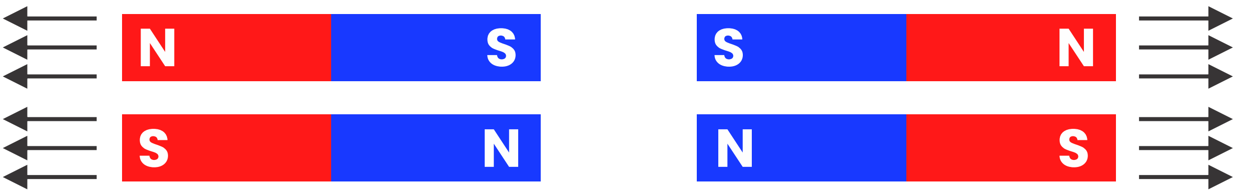
- Materials that get attracted towards a magnet are called magnetic materials. e.g. iron, cobalt or nickel.
- Materials that do not get attracted by a magnet are non-magnetic materials. e.g. wood, plastic etc.

9.3 Properties of magnet

Magnets are objects that produce a magnetic field, enabling them to attract certain materials and interact with other magnets. They possess distinct properties that define their behaviour and applications.

ATTRACTION

REPULSION



1. Attractive Property

- Definition: Magnets attract materials such as iron, nickel, and cobalt.
- Explanation: This attraction occurs because these materials are ferromagnetic, meaning their internal structure allows them to be strongly influenced by a magnetic field.

2. Repulsive Property

- Definition: Like poles of magnets repel each other, while unlike poles attract.
- Explanation: Each magnet has two poles: north and south. When the north pole of one magnet is brought near the north pole of another, they repel each other; the same occurs with two south poles. Conversely, a north pole and a south pole will attract.

3. Magnetic Field

- Definition: The space around a magnet within which magnetic forces are exerted.
- Explanation: This field is invisible but can be visualized using iron filings or a magnet.

9.4 Uses Of Magnets In Daily Life:

1. Household Applications

- Refrigerator Magnets: They are commonly used to close the doors.
- Cabinet Latches: Magnets help keep cabinet doors securely closed, ensuring they remain shut until opened intentionally.

2. Electronics and Technology

- Speakers and Microphones: Magnets convert electrical signals into sound vibrations in speakers and facilitate the reverse in microphones.
- Electric Motors and Generators: Magnets are essential in converting electrical energy to mechanical energy and vice versa, powering various appliances and industrial machines.

3. Medical Field

- Magnetic Resonance Imaging (MRI): MRI machines use powerful magnets to generate detailed images of internal body structures, aiding in medical diagnoses.

5. Security and Navigation

- Magnetic Strips on Cards: Credit and debit cards have magnetic strips that store data, enabling secure financial transactions.

EXERCISES

Fill in the blanks.

1. Materials that are strongly attracted by a magnet are called ______________.

2. ______________ is a metal that is strongly attracted to magnets.

3. The region around a magnet where its influence is felt is called ______________.

5. ______________ is used in electric motors because of its magnetic properties.

6. ______________ are used in magnetic resonance imaging (MRI) machines.

7. ______________ are used to make fridge magnets.

Choose the correct answer

1. Which metal is strongly attracted to magnets?
a) Wood b) Plastic c) Iron d) Glass

2. A freely suspended magnet always points towards:
a) East-West b) North-South c) Up-Down d) Left-Right

3. Which of the following is used in MRI machines?
a) Plastic b) Magnet c) Wood d) Paper

4. The Earth acts like a giant:
a) Balloon b) Magnet c) Stone d) Cloud

Answer the Following Questions:

Short-Answer Questions

1. What is a magnet?
2. Write the difference between magnetic and non-magnetic materials?

Long-Answer Questions

1. List and explain the main properties of magnets.

Short Note

1. Write a short note on uses of magnets in daily life.

ACTIVITY

1. Match the material, which can be attracted by magnet.